

INTERNSHIP MONTHLY REPORT

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Details of the internship schedule:

Key date	Key learning activities
Week 13 (29/9 - 3/10)	Get familiar with the problem of Motion planning, obstacle avoidance and collision prevention
Week 14 (6/10 - 10/10)	Simulation 2d Lidar in obstacle detection and application on UAV
Week 15 (13/10-17/10)	Artificial Potential Field method in obstacle avoidance
Week 16 (20/9 - 31/10)	Summit git, user manual and final documentation.

1. What skills have you learned in the past 4 weeks? Please describe your progress using the suggestions below:

Over the past four weeks, I have developed a solid understanding of motion planning and obstacle avoidance in both mobile robots and unmanned aerial vehicles (UAVs). I explored the fundamental challenges involved in enabling autonomous navigation in dynamic environments, such as path optimization, sensor integration, and real-time decision-making.

I learned about and reviewed existing solutions and algorithms used to achieve effective obstacle avoidance. This included studying classical motion planning algorithms such as A*, Dijkstra's algorithm, and Rapidly Exploring Random Trees (RRT), focusing on how they can be adapted for UAV applications. I analyzed their advantages and disadvantages. for example, how A* and Dijkstra guarantee optimal paths but can be computationally expensive, while RRT is faster and more flexible but may produce suboptimal paths.

In addition, I gained practical knowledge of sensor-based obstacle detection, particularly through the study of 2D LiDAR. I learned how LiDAR functions in mapping and detecting obstacles, its role in Simultaneous Localization and Mapping (SLAM), and how it can be applied in real-time navigation systems. I also explored alternative perception methods, such as computer vision using stereo cameras or depth sensors, comparing their accuracy, processing requirements, and suitability for UAVs with limited payload capacity.

Furthermore, I studied the artificial potential field (APF) method, which models obstacles and goals as virtual forces influencing the robot's motion. I examined how APF can be used for reactive obstacle

avoidance and began experimenting with its implementation in simulation environments to understand its behavior and limitations (e.g., local minima problems).

Overall, these activities have enhanced my skills in robot motion planning, sensor integration, algorithmic analysis, and simulation-based testing. I have developed both theoretical knowledge and practical insight into how different techniques can be combined to enable safe and efficient autonomous navigation for mobile robots and UAVs.

2. Are there any problems or delays with the internship where you need support?

I believed that the internship process ended successfully.

3. With respect to the final report, with questions on professional responsibilities and on consideration of standards and laws:

I have learned to consider professional responsibilities in robotics and UAV development by ensuring safety, reliability, and ethical operation. I studied relevant standards and laws, such as UAV flight regulations, data privacy rules, and safety guidelines, and learned to design systems that comply with these requirements. This has helped me integrate technical solutions like obstacle avoidance and motion planning while ensuring lawful and responsible deployment.